








Modelling Financial Markets

Chapter 5: Models of Conditional Volatility



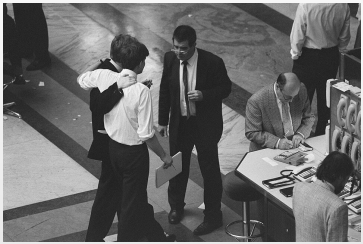
Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Can We Forecast Tomorrow's Risk?

- ARCH means AutoRegressive Conditional Heteroskedasticity: variance depends on past shocks
- Core topics
  - Clustering and the ARCH tests; the GARCH recursion, persistence and half-life; estimation with Student-t innovations; forecasts; asymmetry (GJR, EGARCH); residual diagnostics; forecast evaluation with QLIKE, Mincer–Zarnowitz and Diebold–Mariano
- Inference beyond the basic model
  - QMLE asymptotics, the delta method near $\alpha + \beta = 1$, tests on the boundary, corrected residual tests, Giacomini–White and the Model Confidence Set
- Two research sections: a quasi score-driven volatility model and an AI-assisted discovery loop

Volatility Comes in Bursts

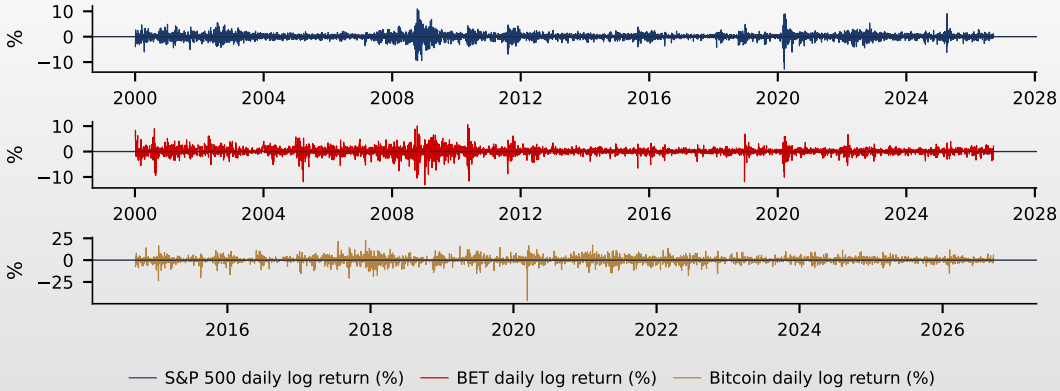


Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- Crashes are not confined to isolated days: large moves are followed by large moves
  - "Large changes tend to be followed by large changes, of either sign" Mandelbrot (1963)
- This **volatility clustering** is stylised fact 6 of Chapter 1
- Volatility enters every risk number: Value at Risk (Chapter 7), portfolio weights (Chapter 4), option prices
- Beyond the forecast: can we conduct valid inference on it?

Volatility Clusters in Every Market



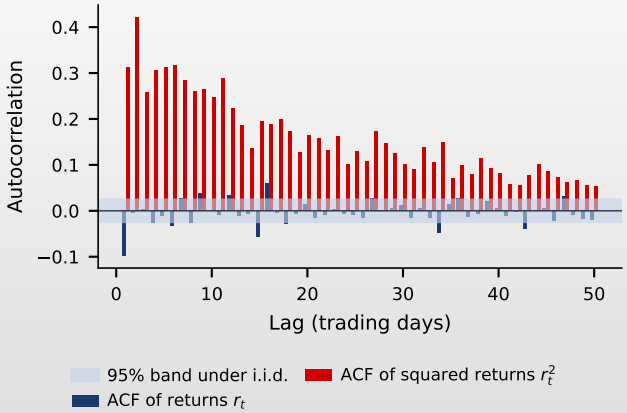
MFM_ch5_arch_effects

Calendar and Volatility Episodes

- Daily log returns $r_t = 100\,(\ln P_t - \ln P_{t-1})$ in %, closes up to 18 September 2026
  - P_t: closing price (index level) on day t
- Six series, chosen for different market regimes: the S&P 500 (the global benchmark) and the BET (the local market), from 2000
- BET-TR (the BET with reinvested dividends), from 2014
- Bitcoin (BTC), from 17 September 2014, 7 days a week
- EUR/RON (the BNR reference rate, a managed float), from 2005; gold (XAU/USD), a safe-haven asset, from 2000
- Equity indices: weekdays; days whose close equals the previous close (holidays filled with the last price) are dropped: 3 for the S&P 500, 0 for the BET
- In the chart (S&P 500, BET, Bitcoin), quiet years alternate with turbulent months: 2008, March 2020, April 2025

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The Sign Is Nearly Unpredictable, the Size Is Not



- Sample autocorrelation at lag k: $\hat{\rho}_k = \widehat{\mathrm{Corr}}(x_t, x_{t-k})$, computed for $x_t = r_t$ and $x_t = r_t^2$
  - Band $\pm 1.96/\sqrt{n}$ (n: number of returns): 95% range of $\hat{\rho}_k$ if the series were i.i.d.
- S&P 500, lag 1: -0.099 for r_t, 0.313 for r_t^2
- For r_t, 15 of 50 autocorrelations lie outside ± 0.024, but all are small; for r_t^2, all 50 do
- Heavy tails too: excess kurtosis (kurtosis minus 3) 10.7 (S&P 500), 11.1 (BET), 12.0 (Bitcoin)
- Object to model: $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$, with $\mathcal{F}_{t-1}$ the information at the close of day $t-1$

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Testing for ARCH Effects (1/2): the Two Tests

- H_0: no clustering, $\mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$ constant; both tests look for autocorrelation in squared returns
- **McLeod–Li** McLeod & Li (1983): the Ljung–Box statistic applied to r_t^2
  - $Q(m) = n(n+2)\sum_{k=1}^{m} \dfrac{\hat{\rho}_k^2}{n-k}$
  - $\hat{\rho}_k$: autocorrelation of r_t^2 at lag k; n: sample size; $m = 10$ lags (two trading weeks)
  - Measures the joint size of the first m autocorrelations; χ^2_{10} under H_0; reject if $Q(10) > 18.3$ (5%)
- **Engle: Lagrange Multiplier (LM) test** Engle (1982): regress $\hat{\varepsilon}_t^2$ on a constant and $\hat{\varepsilon}_{t-1}^2, \ldots, \hat{\varepsilon}_{t-5}^2$ (5 lags: one trading week)
  - $\hat{\varepsilon}_t = r_t - \bar{r}$: demeaned return, $\bar{r}$: sample mean
  - $LM = nR^2$, with R^2 the coefficient of determination of this regression; H_0: all five slopes are 0
  - χ^2_5 under H_0; reject if $LM > 11.1$ (5%); a large LM means past squared shocks predict today's

Testing for ARCH Effects (2/2): Results

- Every statistic is far above its critical value (all $p < 0.001$): clustering in equities, Bitcoin, EUR/RON and gold
- The weakest effect, by $\hat{\rho}_1(r^2)$, is for gold (0.110) and Bitcoin (0.137)
- Q and LM grow with n, so they are not comparable across series of different lengths (one-sided score tests: ▸ Appendix)

Series	$Q(10)$ on r^2	LM(5)	$\hat{\rho}_1(r^2)$
S&P 500	6 133	1 684	0.313
BET	2 440	982	0.330
BET-TR	714	371	0.171
Bitcoin	213	115	0.137
EUR/RON	2 420	871	0.342
Gold	744	329	0.110

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The ARCH Model of Engle (1982)

- Setting: $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, innovations z_t i.i.d. with mean 0 and variance 1
  - μ: mean return; ε_t: shock (unexpected part of the return); σ_t: conditional volatility
  - Conditional variance $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$; unconditional variance $\sigma^2 = \mathbb{E}\sigma_t^2$
- **ARCH**(q): today's variance is a constant plus a weighted sum of the last q squared shocks Engle (1982)
  - $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2$
  - q: number of lags; ω: variance floor; α_i: weight of the shock i days ago
  - $\omega > 0$, $\alpha_i \geq 0$: the variance is always positive
- A large shock yesterday (ε_{t-1}^2 large) raises today's variance: clustering by construction
- ARCH(1): unconditional variance $\sigma^2 = \omega/(1-\alpha_1)$, finite if $\alpha_1 < 1$
- Engle received the 2003 Nobel Prize for ARCH (history: Appendix)

From ARCH to GARCH (Bollerslev, 1986)

- GARCH means Generalised AutoRegressive Conditional Heteroskedasticity: the variance also depends on its own past
- Problem with ARCH: clustering in real data lasts for months, so q must be large (many parameters)
- **GARCH**(1,1) Bollerslev (1986): $\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2$
  - $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$; α: reaction to news; β: memory of past variance
- Substituting backwards: $\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{j\geq 1} \beta^{j-1} \varepsilon_{t-j}^2$
  - Valid for $\beta < 1$; j: age of the shock in days, weight $\alpha\beta^{j-1}$
  - An ARCH(∞) with geometrically declining weights, using only three parameters
- GARCH(p,q) adds more lags (p lags of σ^2, q lags of ε^2); in practice GARCH(1,1) is the workhorse Hansen & Lunde (2005)

Model Labels and Innovations

- The model name describes the variance dynamics; the suffix describes the law of the standardised innovation
- GARCH-N: GARCH with innovations from the Normal distribution
- GARCH-t: GARCH with Student-t innovations, which allow heavier tails
- GJR-t (or GJR-GARCH-t): asymmetric Glosten–Jagannathan–Runkle variance (Asymmetry section) with Student-t innovations

GARCH(1,1) as an ARMA Model for Squared Shocks

- Define $u_t = \varepsilon_t^2 - \sigma_t^2$, the surprise in the squared shock ($\mathbb{E}[u_t \mid \mathcal{F}_{t-1}] = 0$)
- Then $\varepsilon_t^2 = \omega + (\alpha + \beta)\,\varepsilon_{t-1}^2 + u_t - \beta\, u_{t-1}$
  - Obtained by adding u_t to both sides of the GARCH recursion and writing $\sigma_{t-1}^2 = \varepsilon_{t-1}^2 - u_{t-1}$
  - An AutoRegressive Moving Average model, ARMA(1,1), in ε_t^2 with autoregressive coefficient $\alpha + \beta$ and moving-average coefficient $-\beta$
- Consequence: the autocorrelations of ε_t^2 decay geometrically, $\rho_k = \rho_1(\alpha + \beta)^{k-1}$
  - $\rho_k = \mathrm{Corr}(\varepsilon_t^2, \varepsilon_{t-k}^2)$: autocorrelation of squared shocks at lag k
  - $\rho_1 = \alpha(1 - \beta^2 - \alpha\beta)/(1 - \beta^2 - 2\alpha\beta)$ if $\mathbb{E}\varepsilon_t^4 < \infty$ (Seminar 5, A6)
  - S&P 500, GARCH-N by Gaussian QMLE ($\hat{\alpha} = 0.121$, $\hat{\beta} = 0.860$): $\rho_1 = 0.356$ and $\rho_{10} = 0.298$, against the sample values $\hat{\rho}_1 = 0.313$ and $\hat{\rho}_{10} = 0.248$
- $\alpha + \beta$ is therefore called the **persistence** of volatility

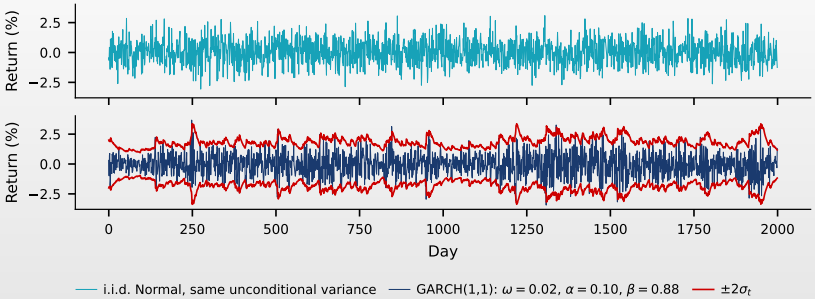
Stationarity and the Unconditional Variance

- **Covariance stationarity**: $\alpha + \beta < 1$; then
  - $\sigma^2 = \mathbb{E}[\sigma_t^2] = \dfrac{\omega}{1 - \alpha - \beta}$
- Derivation: take expectations in the recursion, $\mathbb{E}\sigma_t^2 = \omega + (\alpha + \beta)\,\mathbb{E}\sigma_{t-1}^2$, and set $\mathbb{E}\sigma_t^2 = \mathbb{E}\sigma_{t-1}^2$
- **IGARCH** (integrated GARCH): $\alpha + \beta = 1$; the unconditional variance is infinite
  - Yet, if $\alpha > 0$ and z_t^2 is non-degenerate, the process is still **strictly** stationary: the condition $\mathbb{E}\ln(\alpha z_t^2 + \beta) < 0$ Nelson (1990) holds by Jensen, since $\mathbb{E}\ln(\alpha z_t^2 + \beta) < \ln(\alpha + \beta) = 0$
- Notation: σ^2 is the model's long-run variance (Seminar 5 writes $\bar{\sigma}^2$), $\hat{\sigma}^2 = \hat{\omega}/(1 - \hat{\alpha} - \hat{\beta})$ its estimate; s^2 is the sample variance of r_t
- Variance targeting: fix $\omega = s^2(1 - \alpha - \beta)$, so that the long-run variance equals s^2 and only α, β are estimated

Fourth Moment and Kurtosis

- With Normal z_t, the kurtosis of ε_t is $K = \dfrac{3\left[1-(\alpha+\beta)^2\right]}{1-(\alpha+\beta)^2-2\alpha^2}$
  - $K = \mathbb{E}\varepsilon_t^4/(\mathbb{E}\varepsilon_t^2)^2$: 3 for the Normal distribution, above 3 for heavy tails
  - Finite only if $(\alpha+\beta)^2+2\alpha^2<1$ (Seminar 5, A4)
  - ARCH(1) is the case $\beta=0$: $K=3(1-\alpha^2)/(1-3\alpha^2)$, finite only if $\alpha<1/\sqrt{3}$: heavy tails even with Normal innovations
- S&P 500, GARCH-N, 2000–2026: $\hat{\alpha}=0.120604$, $\hat{\beta}=0.859878$
  - $(\hat{\alpha}+\hat{\beta})^2+2\hat{\alpha}^2=0.990435$: just below 1, so $K=12.1$; sample kurtosis of the returns: 13.7 (excess kurtosis 10.7)
- Near the moment boundary, K is very sensitive to rounding the estimates (Appendix)
- Clustering alone generates much of the fat tails; the rest comes from heavy-tailed z_t

Simulation: GARCH Produces Clustering



- Top: i.i.d. Normal returns; bottom: GARCH(1,1) with $\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$, the same unconditional variance (1), and the bands $\pm 2\sigma_t$
- Quiet stretches and bursts appear only in the GARCH path
- Normal innovations, yet fat-tailed returns: a mixture of Normal distributions with varying variances has heavy tails (kurtosis on a long path: Appendix)

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Persistence and Half-Life

- Multi-step forecast: $\mathbb{E}[\sigma^2_{t+h} \mid \mathcal{F}_t] = \sigma^2 + (\alpha+\beta)^{h-1}(\sigma^2_{t+1} - \sigma^2)$
  - h: forecast horizon in days; σ^2_{t+1}: one-day-ahead variance, known at t
  - The gap to the long-run variance shrinks by the factor $\alpha+\beta$ each day
- **Half-life**: number of days after which half of a variance shock has gone
  - $(\alpha+\beta)^h = 0.5 \;\Rightarrow\; h_{1/2} = \dfrac{\ln 0.5}{\ln(\alpha+\beta)}$
- Half-life: 13.5 days for $\alpha+\beta = 0.95$; 69 days for 0.99; 138 days for 0.995
- Small changes in $\alpha+\beta$ near 1 imply very different half-lives: estimation uncertainty matters (next section; Seminar 5, A2 and B4)

A Toy Check: Multi-Day Risk After a Turbulent Day

- Toy GARCH(1,1), not fitted to any data: $\omega = 0.04$, $\alpha = 0.08$, $\beta = 0.90$
  - Persistence $p = \alpha + \beta = 0.98$; long-run variance $\sigma^2 = 0.04/0.02 = 2$; half-life $\ln 0.5 / \ln 0.98 \approx 34.3$ days
- A turbulent day: $\sigma^2_{t+1} = 4$, twice the long-run level
  - Day 2: $\mathbb{E}[\sigma^2_{t+2} \mid \mathcal{F}_t] = 2 + 0.98 \times (4 - 2) = 3.96$: the gap to σ^2 shrinks by 2% a day
- H-day variance: daily shocks are uncorrelated, so the variance of their sum is the sum of the daily variance forecasts, a geometric sum
  - $V_H = \sum_{h=1}^{H} \mathbb{E}[\sigma^2_{t+h} \mid \mathcal{F}_t] = H\sigma^2 + (\sigma^2_{t+1} - \sigma^2)\,\dfrac{1 - p^H}{1 - p}$
  - $H = 5$: $V_5 = 10 + 2 \times (1 - 0.98^5)/0.02 = 10 + 2 \times 4.804 = 19.61$; the square-root-of-time rule, $V_H = H\,\sigma^2_{t+1}$ (H-day volatility is $\sqrt{H}$ times the one-day volatility), gives $5 \times 4 = 20$, too high
- Your turn: redo $H = 5$ after a calm day, $\sigma^2_{t+1} = 1$ (answer: $V_5 = 10 - 4.804 = 5.20 > 5$, so the rule is now too low, as in the S&P 500 case of the Forecasts section)

EWMA and IGARCH: No Mean Reversion

- **Exponentially Weighted Moving Average (EWMA)** J.P. Morgan/Reuters (1996): $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1-\lambda) r_{t-1}^2$, with $\lambda = 0.94$ for daily data and zero-mean returns
  - $\lambda \in (0,1)$: decay factor
  - Weights $(1-\lambda)\lambda^{j-1}$ on past squared returns: the most recent days count most
  - EWMA is an IGARCH(1,1) with $\omega = 0$, $\alpha = 1-\lambda$, $\beta = \lambda$: no parameter to estimate
- Multi-step forecasts at the close of day t differ
  - $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot \mid \mathcal{F}_t]$: expectation given the information at t
  - EWMA: flat, $\mathbb{E}_t \sigma_{t+h}^2 = \sigma_{t+1}^2$ for every h
  - IGARCH with $\omega > 0$: rising, $\mathbb{E}_t \sigma_{t+h}^2 = \sigma_{t+1}^2 + (h-1)\omega$
  - Stationary GARCH: pulled towards σ^2 at the rate $\alpha + \beta$
- EWMA serves as a benchmark in the forecast comparison

Maximum Likelihood

- Maximum Likelihood Estimation (MLE) of $\theta = (\mu, \omega, \alpha, \beta)$; given θ, the recursion produces $\sigma_1^2(\theta), \ldots, \sigma_n^2(\theta)$
- Normal log-likelihood: $\ell(\theta) = -\frac{1}{2} \sum_{t=1}^{n} \left[\ln 2\pi + \ln \sigma_t^2 + \varepsilon_t^2 / \sigma_t^2 \right]$
  - $\varepsilon_t = r_t - \mu$
  - Each day contributes the density of ε_t given the past: the likelihood factorises in time
- No closed form: numerical optimisation under $\omega > 0$, $\alpha, \beta \geq 0$, $\alpha + \beta \leq 1$
  - The `arch` package imposes $\alpha + \beta \leq 1$, a closed set: an estimate can sit on the bound (Bitcoin and EUR/RON in the six-market table)
- Start-up: σ_1^2 is a backcast from the first squared residuals, whose effect fades at rate β
- Scale: returns in %, so that the variance is of order 1
  - EUR/RON has a daily standard deviation of about 0.28%, so it is fitted on 10 r_t; on unscaled returns the optimiser stops at a local optimum

Heavy-Tailed Innovations

- **Student-t** Bollerslev (1987): $z_t = \sqrt{(\nu-2)/\nu}\, t_\nu$, unit variance for $\nu > 2$
  - t_ν: a standard Student-t variable, variance $\nu/(\nu-2)$; the factor rescales it to variance 1
  - ν (degrees of freedom): the smaller ν, the heavier the tails; the fourth moment exists only for $\nu > 4$
- **Skewed-t** Hansen (1994): adds an asymmetry parameter $\lambda \in (-1, 1)$
  - $\lambda < 0$: the left tail is longer (large losses more likely than large gains)
- **Generalised Error Distribution (GED)** Nelson (1991): density $\propto \exp(-\frac{1}{2}|z/\lambda_\nu|^\nu)$
  - $\nu > 0$: shape parameter; λ_ν: scale constant
  - $\nu = 2$: the Normal distribution; $\nu = 1$: Laplace; $\nu < 2$: heavier tails (λ_ν gives unit variance)

The Student-t Likelihood

- Day t's contribution to the GARCH-t log-likelihood, the chapter's baseline model ($\varepsilon_t = r_t - \mu$)

$$\ell_t = \ln \Gamma\left(\tfrac{\nu+1}{2}\right) - \ln \Gamma\left(\tfrac{\nu}{2}\right) - \tfrac{1}{2} \ln[\pi(\nu-2)] - \tfrac{1}{2} \ln \sigma_t^2 - \tfrac{\nu+1}{2} \ln\left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2}\right)$$

- Γ: the gamma function; $\nu > 2$: degrees of freedom of the unit-variance innovation
- As $\nu \to \infty$, ℓ_t becomes the Normal contribution of the Maximum Likelihood slide
- Why Student-t: a single extra parameter (ν) and symmetric heavy tails

Gaussian QMLE (1/2): Definition and Consistency

- **Gaussian QMLE** (Quasi-Maximum Likelihood Estimator): maximise the Normal likelihood even if z_t is not Normal Bollerslev & Wooldridge (1992)
  - The Normal $\ell(\theta)$ is used as a criterion, not as a claim about the law of z_t
  - θ_0: true parameter value; $\hat{\theta}$: the estimate, the maximiser of ℓ
- **Consistency** (GARCH(1,1)): $\hat{\theta} \to \theta_0$ as $n \to \infty$ Berkes, Horváth & Kokoszka (2003), Francq & Zakoïan (2004)
  - Strict stationarity: $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$
  - $\alpha_0 > 0$ (otherwise β is not identified) and $\beta_0 < 1$
  - θ_0 in a compact (closed and bounded) parameter set; z_t^2 non-degenerate (not a constant)
  - No moment condition on ε_t: consistency holds even for IGARCH

Gaussian QMLE (2/2): Asymptotic Normality and Standard Errors

- **Asymptotic normality** of $\vartheta = (\omega, \alpha, \beta)$ (pure GARCH, known mean): add ϑ_0 interior and $\kappa_z = \mathbb{E}z_t^4 < \infty$
  - $\sqrt{n}(\hat{\vartheta} - \vartheta_0) \to N\left(0, (\kappa_z - 1)J^{-1}\right)$, $J = \mathbb{E}[g_t g_t']$, $g_t = \partial \ln \sigma_t^2 / \partial \vartheta$
  - κ_z: kurtosis of z_t (3 if Normal); g_t: gradient of $\ln \sigma_t^2$; J: its mean outer product; interior: no constraint binds
- Sandwich $\hat{A}^{-1}\hat{B}\hat{A}^{-1}$ estimates $(\kappa_z - 1)J^{-1}$; the inverse Hessian $\hat{A}^{-1}$ estimates $2J^{-1}$
  - A: minus the expected Hessian (second derivatives) of ℓ/n; B: variance of the score (first derivatives) per observation
  - Robust / classical standard error (SE) $= \sqrt{(\kappa_z - 1)/2}$: 1 under Normality, above 1 with heavy tails
- Variants (estimated mean, IGARCH, explosive case): ▸ Appendix

QMLE in Our Data: When the Theory Holds and When It Fails

- S&P 500, GARCH(1,1) with Normal likelihood: residual kurtosis $\hat{\kappa}_z = 4.73$
  - Theory: robust / classical SE $= \sqrt{(4.73-1)/2} = 1.37$ (Seminar 5, A5); observed: 1.37 for $\hat{\alpha}$, 1.37 for $\hat{\beta}$
- Bitcoin: GARCH-t gives $\hat{\nu} = 3.18 < 4$ degrees of freedom: under that law $\mathbb{E}z_t^4 = \infty$, and the condition for Gaussian-QMLE normality fails
  - Gaussian QMLE residual kurtosis does not settle: 19.8 on the first half, 14.6 on the full sample
  - Full sample, robust / classical SE: 2.51 for $\hat{\alpha}$, 2.01 for $\hat{\beta}$, against the formula value 2.61, which assumes $\kappa_z < \infty$
- Hall & Yao (2003): then the Gaussian QMLE converges slower than $\sqrt{n}$, to a non-Gaussian limit: its robust t-statistics are not valid
- Remedies: a heavy-tailed quasi-likelihood with its scale adjustment Fan, Qi & Xiu (2014), or the Student-t MLE (bounded score: no fourth moment needed if the law is right), with bootstrap inference near bounds

MFM_ch5_inference

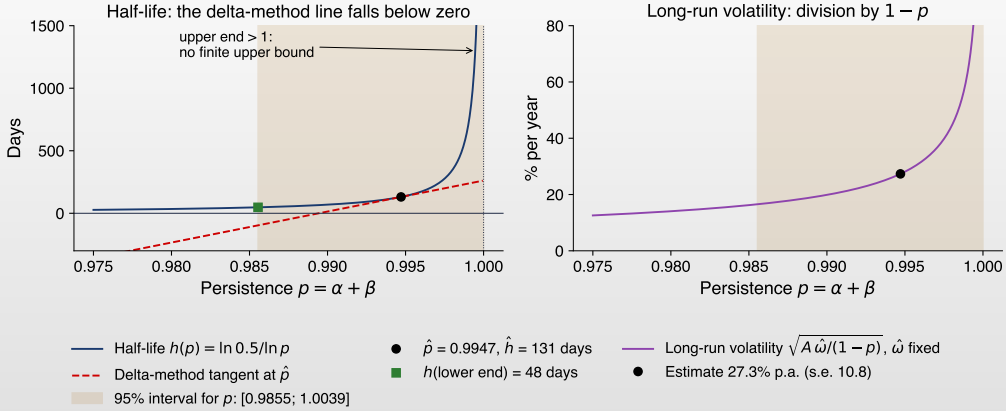
S&P 500: GARCH(1,1)-t Estimates

Parameter	Estimate	Robust SE	Classical SE	Ratio
ω	0.0157			
α	0.123	0.0107	0.0101	1.06
β	0.872	0.0103	0.0096	1.08
ν	6.26	0.48		
$\alpha + \beta$	0.9947			

- Mean $\hat{\mu} = 0.079\%$ per day; full table with every standard error: Seminar 5, B1
- Large $\hat{\beta}$: the variance memory dominates; $\hat{\alpha}$: a moderate news reaction; $\hat{\nu} = 6.26$: tails much heavier than Normal
- Half-life 131 days; long-run volatility $\sqrt{251.4\,\hat{\omega}/(1-\hat{\alpha}-\hat{\beta})} = 27.3\%$ p.a. ($A = 251.4$ returns a year), above the sample 19.2%: dividing by $1-\hat{\alpha}-\hat{\beta}$ makes it imprecise (next slide)

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Near-Unit Persistence: What the Delta Method Can and Cannot Do (1/2)



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Near-Unit Persistence: What the Delta Method Can and Cannot Do (2/2)

- $g(\theta)$: a smooth function of the parameters, e.g. the long-run volatility or the half-life; ∇g: its gradient
- Delta method: $\widehat{\mathrm{SE}}\, g(\hat{\theta}) \approx \sqrt{\nabla g' \hat{V} \nabla g}$, with $\hat{V}$ the robust covariance of $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$
- For the half-life: $p = \alpha + \beta$ and $h = \ln 0.5 / \ln p$
- Long-run volatility 27.3% p.a., SE 10.8 percentage points
- Half-life: $\partial h / \partial p = 24\,789$ at $\hat{p} = 0.994712$, $\widehat{\mathrm{SE}}(\hat{p}) = 0.0047$, so $\widehat{\mathrm{SE}}(\hat{h}) \approx 116$ days, and $\hat{h} \pm 1.96\, \widehat{\mathrm{SE}}$ includes negative half-lives: the linearisation fails
- Transform the interval for p instead: $[0.9855; 1.0039]$ gives $h \in [48; \infty)$; whether $p = 1$ is plausible needs a boundary test (next slides)

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Testing on the Boundary (1/2): a Decision Table

Situation under H_0	**Example**	**Likelihood Ratio (LR):** $LR = 2(\ell_1 - \ell_0)$	**5% critical value**
Interior restriction	GJR (asymmetric GARCH) vs GARCH, $\gamma = 0$ (with $\alpha > 0$)	χ^2_1	3.84
One parameter on the boundary, the others identified and interior	Normal vs Student-t ($1/\nu = 0$); IGARCH ($\alpha + \beta = 1$)	$\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ Self & Liang (1987), Andrews (2001)	2.71
A nuisance parameter unidentified	No ARCH: $\alpha = 0$ leaves β unidentified	non-standard Andrews (2001)	simulate
Several bounds active at once	GJR with $\hat{\alpha} = 0$, test of $\gamma = 0$	non-standard	simulate

- ℓ_1, ℓ_0: maximised log-likelihoods without and with the restriction; $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$: $LR = 0$ with probability $\frac{1}{2}$, otherwise χ^2_1
- GJR: γ is the extra variance reaction to negative shocks (Asymmetry section)

Testing on the Boundary (2/2): Why IGARCH Is a Boundary Case

- IGARCH is on the boundary only because the fit restricts $\alpha + \beta \leq 1$, as `arch` does
  - Under $H_0 : \alpha + \beta = 1$ the unrestricted estimate would exceed 1 about half of the time; the restriction sets it to 1 and then $LR = 0$: hence the mixture and 2.71
  - Without that restriction, $\alpha + \beta = 1$ lies inside the strictly stationary region, the QMLE is still asymptotically Normal there Lumsdaine (1996), and the LR would be compared with χ^2_1 (3.84)
- A Wald t for an estimate at the bound (e.g. $\hat{\alpha} = 0$) is not $N(0, 1)$; a parametric bootstrap from $\hat{\theta}$ is inconsistent when θ_0 is on the boundary Andrews (2000): simulate under H_0 instead

Boundary Tests on Our Data

- Normal vs Student-t innovations, GARCH(1,1): $LR = 311.0$ (S&P 500), 1 082.0 (Bitcoin; Seminar 5, B6)
  - Far above 2.71: heavy tails are decisive; the mixture matters only in borderline cases
- IGARCH, $H_0 : \alpha + \beta = 1$ (GARCH-t): $LR = 1.11$ (S&P 500), 1.14 (BET)
  - S&P 500, p-value: 0.29 (χ^2_1), 0.15 (mixture), 0.17 (parametric bootstrap under H_0 on $\hat{\alpha} + \hat{\beta}$, $B = 999$ simulated samples)
  - BET, p-value: 0.28, 0.14 and 0.17 (Seminar 5, B8)
- S&P 500, bootstrap under H_0: 52% of the re-estimates have $\hat{\alpha} + \hat{\beta} \geq 0.999$ (at the bound $\alpha + \beta = 1$), as the mixture predicts; conclusion: a unit root in variance cannot be rejected for the S&P 500 or the BET

MFM_ch5_seminar

Worked Example (1/2): The One-Day Forecast

- S&P 500, GARCH(1,1)-t: $\hat{\omega} = 0.0157$, $\hat{\alpha} = 0.123$, $\hat{\beta} = 0.872$, $\hat{\mu} = 0.079$
  - Persistence $\hat{\rho} = \hat{\alpha} + \hat{\beta} = 0.994712$
  - With the rounded values shown, $1 - \hat{\rho} = 0.005$ and the long-run variance would be 3.14 instead of 2.97
- Last day $T = 18.09.2026$: $\hat{\varepsilon}_T = r_T - \hat{\mu} = 0.088$, filtered variance $\hat{\sigma}^2_T = 0.565$
  - Filtered variance: the recursion run through the whole sample with $\hat{\theta}$, read off on the last day T
  - $\hat{\sigma}^2_{T+1} = \hat{\omega} + \hat{\alpha}\hat{\varepsilon}^2_T + \hat{\beta}\hat{\sigma}^2_T = \hat{\omega} + 0.0009 + 0.493 = 0.509$, i.e. 11.3% p.a.
- A quiet last day: $\hat{\sigma}^2_{T+1}$ is far below the long-run level, so the forecasts will rise (next slide)

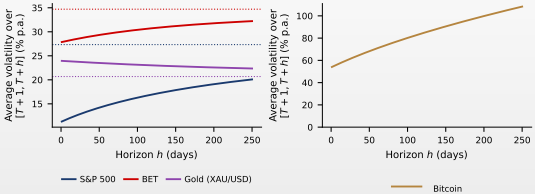
MFM_ch5_garch_estimation

Worked Example (2/2): Ten Days Ahead

- Long-run variance $\hat{\sigma}^2 = \hat{\omega}/(1-\hat{\alpha}-\hat{\beta}) = 2.97$; day 10: $\hat{\sigma}^2 + (\hat{\alpha}+\hat{\beta})^9(\hat{\sigma}^2_{T+1} - \hat{\sigma}^2)$
  - $= 2.97 + 0.953 \times (0.509 - 2.97) = 0.624$
- 10-day variance $\sum_{h=1}^{10} \hat{\sigma}^2_{T+h} = 5.67$, volatility 2.38%
  - The sum in closed form (toy check in the persistence section): $10\,\hat{\sigma}^2 + (\hat{\sigma}^2_{T+1} - \hat{\sigma}^2)\,(1-\hat{\rho}^{10})/(1-\hat{\rho})$
  - $= 29.72 + (0.509 - 2.972) \times 9.765 = 5.67$
  - The square-root-of-time rule gives $\sqrt{10} \times 0.714 = 2.26\%$: too low, because $\hat{\sigma}^2_{T+1} < \hat{\sigma}^2$ and volatility is forecast to rise

MFM_ch5_garch_estimation

Volatility Term Structure



- Average volatility over the next h days, forecast from 18.09.2026 (GARCH(1,1)-t); dotted: long-run level
- S&P 500: from 11.3% for the next day to 20.1% on average over 250 days (below its long run today); gold falls from 23.9% to 22.4%
- Bitcoin and EUR/RON: the estimate hits the bound $\hat{\alpha} + \hat{\beta} = 1$ (IGARCH) with $\hat{\omega} > 0$ (for EUR/RON, $\hat{\omega} \approx 5 \cdot 10^{-8}$ in $\%^2$), so the variance forecast grows linearly by $\hat{\omega}$ per day
- Bitcoin: 54.1% for the next day, 108.3% on average over 250 days (about 8 calendar months); EUR/RON is not in the chart
- This unbounded growth is not a usable forecast: in Chapter 8, GARCH forecasts are withheld on days when the fit is not admissible (Bitcoin: 1 660 of 3 384 evaluation days; EUR/RON: 20 of 4 352)

MFM_ch5_persistence

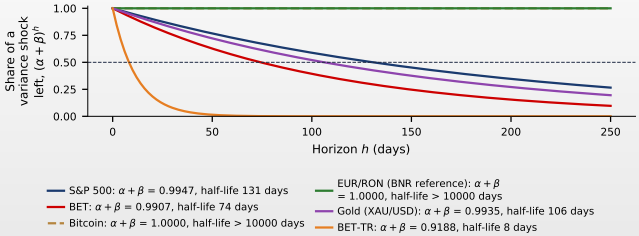
GARCH(1,1)-t in Six Markets

Series	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha}+\hat{\beta}$	$h_{1/2}$ (days)	$\hat{\nu}$	Long-run vol.
S&P 500	0.0157	0.123	0.872	0.9947	131	6.26	27.3
BET	0.0448	0.192	0.799	0.9907	74	5.01	34.7
BET-TR	0.0755	0.206	0.713	0.9188	8	5.02	15.2
Bitcoin	0.1937	0.109	0.891	1.0000	∞	3.18	–
EUR/RON	0.0000	0.123	0.877	1.0000	∞	4.20	–
Gold	0.0107	0.049	0.944	0.9935	106	4.58	20.7

- Long-run volatility in % p.a., each series annualised with its own frequency; – when $\hat{\alpha}+\hat{\beta} \geq 0.999$; EUR/RON: $\hat{\omega}$ in $\%^2$, after the fit on $10\,r_t$
- S&P 500, BET and gold: persistence 0.99–0.995; the BET reacts more to news ($\hat{\alpha} = 0.192$) and forgets faster than the S&P 500
- BET-TR covers only 2014–2026: a comparison with the BET needs the same sample
- EUR/RON: on 6 May 2025 a move of $+1.20\%$ after days of $\pm 0.004\%$ gives a standardised residual of 154

MFM_ch5_garch_estimation

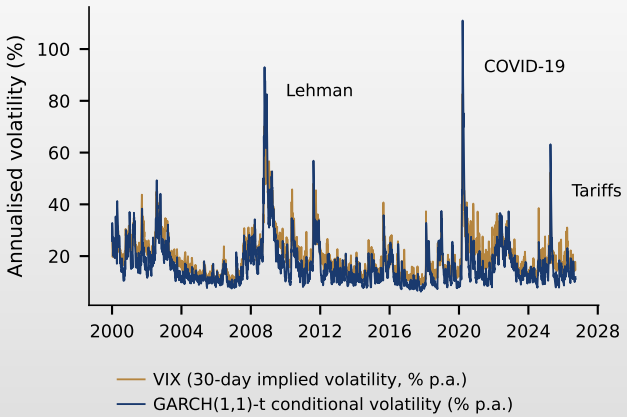
Persistence: How Fast Do Shocks Fade?



- $(\hat{\alpha}+\hat{\beta})^h$: the share of a variance shock left after h days; dashed line: one half
- Half-lives: column $h_{1/2}$ of the six-market table; in Bitcoin and EUR/RON shocks never fade in the model
- Level shifts in volatility can mimic high persistence Lamoureux & Lastrapes (1990) (regime switching: Hamilton & Susmel (1994)): report half-lives as intervals (Seminar 5, B4)

MFM_ch5_persistence

S&P 500: GARCH Volatility and the VIX (1/2)



- $\hat{\sigma}_t\sqrt{A}$: annualised GARCH(1,1)-t volatility
- VIX: 30-day implied volatility (Volatility Index of the Chicago Board Options Exchange)
- Chart labels: Lehman (Lehman Brothers bankruptcy, September 2008), COVID-19 (Coronavirus Disease 2019, March 2020), Tariffs (US tariff announcement, April 2025)

MFM_ch5_sp500_vix

S&P 500: GARCH Volatility and the VIX (2/2)

- Correlation of the two series, 2000–2026: 0.89
- $\hat{\sigma}_t$ is filtered with parameters estimated on the full sample (in-sample), not an out-of-sample forecast
- The VIX is higher on average: 19.8% against 17.0%; the VIX exceeds GARCH volatility on 84% of the days
- Not a measured variance risk premium: the premium compares VIX^2 with a 30-day variance forecast under the real-world probability; $\hat{\sigma}_t$ is a one-day volatility

MFM_ch5_sp500_vix

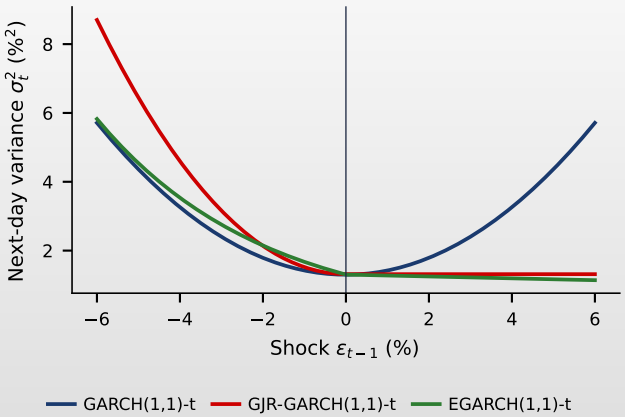
GJR-GARCH (Glosten, Jagannathan and Runkle, 1993)

- **The question**: does volatility rise more after a fall than after a rise of the same size (stylised fact 9, Chapter 1)?
- GARCH(1,1) cannot see the sign: ε^{2}_{t-1} is the same for -3 and $+3$
- GJR: a negative shock gets the extra slope γ Glosten, Jagannathan & Runkle (1993)
  - $\sigma^{2}_{t} = \omega + (\alpha + \gamma\, I_{t-1})\,\varepsilon^{2}_{t-1} + \beta\,\sigma^{2}_{t-1}$, with $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$, else 0
  - Good news: slope α; bad news: slope $\alpha + \gamma$; positivity: $\omega > 0$, $\alpha \geq 0$, $\alpha + \gamma \geq 0$, $\beta \geq 0$ (γ may be negative)
  - Persistence $\alpha + \gamma/2 + \beta$ for symmetric z_t; for skewed z_t replace $1/2$ by $\mathbb{E}[z_t^2 1(z_t < 0)]$, $1(\cdot)$ the indicator (Seminar 5, A3)
- S&P 500: $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ (t = 10.0), $\hat{\beta} = 0.881$; symmetry test: $H_0 : \gamma = 0$ by t or by LR against GARCH(1,1)
- $\hat{\alpha} = 0$ is on the boundary: the t-test of γ is not standard, and the χ^{2}_{1} law of the LR for $\gamma = 0$ assumes $\alpha_0 > 0$ (decision table for tests on the boundary)
- Candidate mechanisms, not tested here: leverage (a falling price raises the debt-to-equity ratio) and volatility feedback (higher expected volatility raises required returns, so prices fall now)

EGARCH (Nelson, 1991)

- Exponential GARCH (EGARCH) models the log of the variance, driven by the size and the sign of yesterday's standardised shock Nelson (1991)
  - $\ln \sigma_t^2 = \omega + \alpha\left(|z_{t-1}| - \mathbb{E}|z_{t-1}|\right) + \gamma\, z_{t-1} + \beta \ln \sigma_{t-1}^2$
  - $z_{t-1} = \varepsilon_{t-1}/\sigma_{t-1}$: standardised shock; $|z_{t-1}| - \mathbb{E}|z_{t-1}|$: its size above the average size
  - Size effect α, sign effect γ; $\gamma < 0$: negative shocks raise volatility more
- The log form keeps $\sigma_t^2 > 0$ without any sign restriction on the parameters
- Persistence: β (stationary if $|\beta| < 1$)
- S&P 500: $\hat{\alpha} = 0.138$, $\hat{\gamma} = -0.165$ (t $= -15.2$), $\hat{\beta} = 0.979$
- In `arch`, $\mathbb{E}|z|$ is set to $\sqrt{2/\pi}$, its value under the Normal distribution

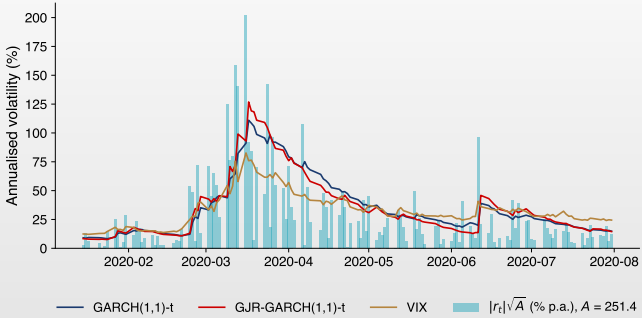
The News Impact Curve for the S&P 500



- **News impact curve** Engle & Ng (1993): σ_t^2 as a function of ε_{t-1}, with σ_{t-1}^2 fixed at the sample variance $s^2 = 1.47$ (%^{2}), not at the long-run variance of the model
- Shock of -3%: GJR 3.16, EGARCH 2.76; $+3$%: GJR 1.31, EGARCH 1.22; GARCH: 2.40 for both
- After bad news the forecast is 2.4 times (GJR) and 2.3 times (EGARCH) that after good news: ratios of levels, not of increases

MFM_ch5_asymmetry

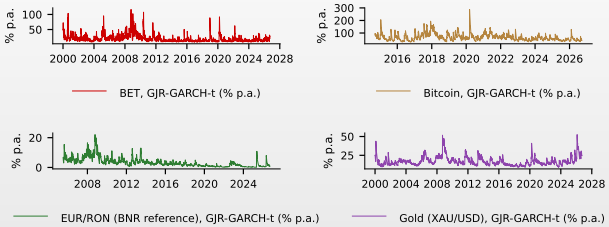
Zoom: The COVID-19 Crash, 2020



- COVID-19, January–July 2020: $|r_t|$ annualised, GARCH-t and GJR-GARCH-t volatility, VIX
- 24 February ($r_t = -3.41\%$): the VIX went from 17.1% to 25.0% the same day; GARCH-t from 12.2% to 22.5% only the next day: GARCH reacts, markets price at once
- After $r_t = -12.8\%$ on 16 March, peaks on 17 March: GARCH-t 111%, GJR-t 127%: the GJR peak is higher because the shocks were negative; full-sample (2000–2026) parameters; a forecast with parameters estimated before 2020 (as in Seminar 5, B5) gives a different peak

MFM_ch5_sp500_vix

BET, Bitcoin, EUR/RON and Gold



- BET: 2008 dominates, now 26.4%; Bitcoin: 50–100% is normal, the highest peak is March 2020
- EUR/RON: a managed calm with bursts in 2022 and 2025, poorly described by a daily GARCH; gold: a new high of 53% in 2026
- Bitcoin and EUR/RON: the fit is on the IGARCH bound, so the paths are descriptive only (term structure)

MFM_ch5_markets

Asymmetry in Six Markets

Series	$\hat{\gamma}_{GJR}$	t	LR vs GARCH	p-value	$\hat{\gamma}_{EGARCH}$	t
S&P 500	0.205	10.0	235.7	< 0.001	-0.165	-15.2
BET	0.043	2.1	4.9	0.026	-0.028	-2.6
BET-TR	0.111	2.9	10.3	0.001	-0.063	-3.5
Bitcoin	-0.013	-0.7	0.7	0.416	0.015	1.2
EUR/RON	0.039	3.0	9.3	0.002	-0.040	-2.5
Gold	-0.043	-4.7	27.6	< 0.001	0.038	5.7

- Student-t innovations; robust t; $LR \sim \chi^2_1$ (3.84) if the Student-t density is correct; the robust t is $N(0,1)$ only if $\mathbb{E}z_t^4 < \infty$ (not for Bitcoin, nor for EUR/RON with EGARCH: $\hat{\nu} < 4$) and the other parameters are interior (not for the S&P 500 GJR, where $\hat{\alpha} = 0$)
- Bad news weighs more (GJR $\hat{\gamma} > 0$, EGARCH $\hat{\gamma} < 0$): S&P 500 strongly; smaller but significant at 5% for the BET, BET-TR and EUR/RON; Bitcoin: none detected; gold: inverted sign
- Leverage for equities and a safe-haven role for gold are interpretations consistent with these signs, not tested mechanisms

MFM_ch5_asymmetry

S&P 500: Twelve Models Compared by BIC

ΔBIC	$N(0,1)$	Student-t	Skewed-t	GED
GARCH(1,1)	648.9	346.7	307.4	339.8
GJR-GARCH(1,1)	394.7	119.9	49.6	141.6
EGARCH(1,1)	372.9	74.8	0.0	111.4

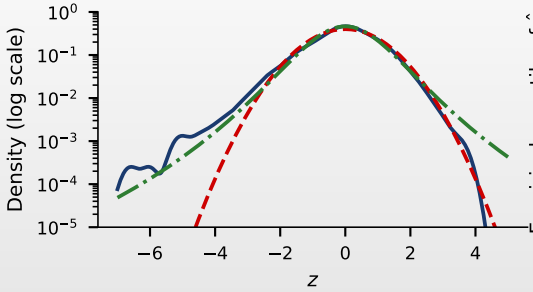
- ΔBIC: Bayesian Information Criterion, BIC $= -2\ell + k \ln n$ (k parameters), minus the lowest BIC among the 12 models (0 = preferred model)
- Akaike Information Criterion, AIC $= -2\ell + 2k$: we choose by BIC, which with $n = 6\,714$ ($\ln n \approx 8.8$) penalises each parameter 4.4 times more and avoids over-parameterised models
- Two choices matter: Student-t instead of Normal innovations lowers BIC by 275–302; asymmetry lowers it by a further 226.8–271.9 (Student-t column, GJR and EGARCH against GARCH); winner: EGARCH with skewed-t innovations
- BIC ranks in-sample fit, not forecast accuracy: the forecast section tests whether these gains survive out-of-sample
- Seminar 5, B3 re-estimates the BET row of the asymmetry table and explains the LR, AIC and BIC decisions

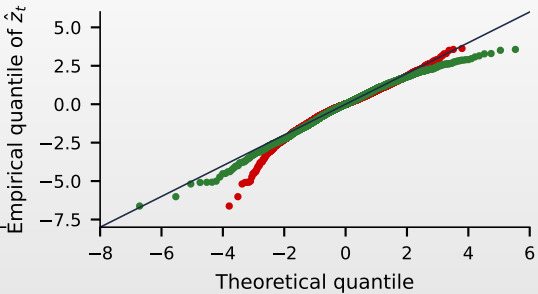
MFM_ch5_garch_estimation

Standardised Residuals: What We Check

- **The question**: has the chosen model removed all dynamics from $\hat{z}_t$, and does $\hat{z}_t$ have the assumed density?
- **Standardised residual**: $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$; if the model is right, $\hat{z}_t$ is close to i.i.d. with variance 1
- Checks, each with H_0: the model captures that feature
  - Mean equation: Ljung–Box $Q(10)$ on $\hat{z}_t$; variance equation: $Q(10)$ on $\hat{z}_t^2$ and LM(5); distribution: Quantile–Quantile (QQ) plot against the assumed density
- Sign-bias regression Engle & Ng (1993): $\hat{z}_t^2 = b_0 + b_1 S_{t-1}^- + b_2 S_{t-1}^- \hat{\varepsilon}_{t-1} + b_3 S_{t-1}^+ \hat{\varepsilon}_{t-1} + u_t$
  - $S_{t-1}^- = 1$ if $\hat{\varepsilon}_{t-1} < 0$, else 0; $S_{t-1}^+ = 1 - S_{t-1}^-$; u_t: regression error; the size terms use the unstandardised residual $\hat{\varepsilon}_{t-1}$ (eq. 18 in Engle & Ng (1993))
  - b_1 (sign bias): $\hat{z}_t^2$ larger on average after a negative shock, and the sign-bias test is the t of b_1; b_2, b_3: negative and positive size bias; joint test $nR^2 \sim \chi_3^2$ (7.81 at 5%), conservative because it omits the gradient of the fitted variance

S&P 500: Are the Innovations Normal? (1/2)





Standardised residuals $\hat{z}_t$ (kernel density)
Normal N(0,1)
Student-t, $\nu = 6.3$ (unit variance)

vs Normal quantiles
vs Student-t ($\nu = 6.3$) quantiles

MFM_ch5_innovations

S&P 500: Are the Innovations Normal? (2/2)

- Standardised residuals of GARCH(1,1) with Normal innovations: excess kurtosis falls from 10.7 (returns) to 1.73; skewness -0.53, Jarque–Bera 1 153
- Jarque–Bera $= n[S^2/6 + E^2/24]$, S: skewness, E: excess kurtosis; χ^2_2 under Normality, 5% critical value 5.99: Normality is rejected by a wide margin
- 15 residuals below -4: 0.21 expected under the Normal, 8.5 under a Student-t with $\nu = 6.3$; the smallest standardised residual: $\hat{z} = -6.6$, on 27 February 2007
- A longer left tail: skewed-t with $\hat{\lambda} = -0.155$ (EGARCH, $\hat{\nu} = 7.5$ degrees of freedom), hence BIC prefers skewed-t innovations

MFM_ch5_innovations

S&P 500: Diagnostics of Five Models

Model	$Q(10)$ on $\hat{z}$	$Q(10)$ on $\hat{z}^2$	LM(5)	Sign bias t	Joint χ^2_3
GARCH-N	21.2	16.4	8.0	3.64	41.8
GARCH-t	21.9	13.6	5.8	3.56	40.7
GJR-t	22.3	10.2	4.5	3.61	28.1
EGARCH-t	27.6	10.5	8.2	2.04	9.9
GJR-skewed-t	19.7	10.7	5.0	3.91	31.8

- 5% critical values: 18.3 (χ^2_{10}), 11.1 (χ^2_5), 7.81 (χ^2_3), $|t| > 1.96$; the χ^2 critical values ignore parameter estimation, so the tests are conservative (next slide)
- Mean: $Q(10)$ on $\hat{z}$ between 19.7 and 27.6 in every model, above the critical value 18.3: a little autocorrelation that a constant mean ignores
- Variance: $Q(10)$ on $\hat{z}^2$ falls from 6 133 (on r_t^2) to 10–16 and LM(5) is not significant: these naive tests detect no clustering left
- Asymmetry: the joint test rejects all five models at 5%: GARCH-t 40.7, GJR-t 28.1, EGARCH-t 9.9 (p-value 0.020, the smallest statistic)
- The sign-bias test (t of b_1) stays above 1.96 in every model (2.04–3.91); with $n = 6\,714$, judge economic size, not only p-values

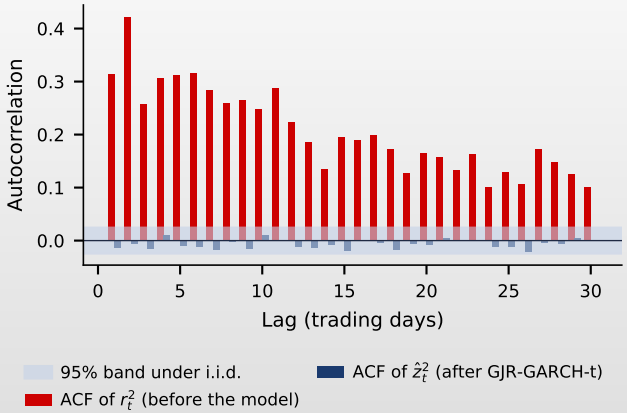
MFM_ch5_innovations

Residual Diagnostics Have Estimated Parameters

- $\hat{z}_t$ depends on $\hat{\theta}$: the Ljung–Box statistic on $\hat{z}_t^2$ is not exactly χ^2_m Li & Mak (1994); the fitted variance has already absorbed part of the dependence at small lags
- S&P 500, Gaussian QMLE of GARCH(1,1), $Q(10)$ on $\hat{z}_t^2$
  - Naive Ljung–Box: 16.4, p-value 0.089: not rejected at 5%
  - Corrected for the estimated parameters: 19.1, p-value 0.038: rejected at 5%, the verdict flips
- The naive test is conservative: it rejects a misspecified variance equation too rarely; the joint sign-bias test carries the same caveat (the correction matrix: Appendix)

MFM_ch5_inference

Diagnostics: Squared Returns versus Squared Residuals



- S&P 500, lags 1–30: autocorrelations of r_t^2 (before the model) and of $\hat{z}_t^2$ from GJR-GARCH-t
- After the model they sit inside the i.i.d. band: no remaining variance dynamics are detected at these lags
- The band is the naive one: with estimated parameters it is too wide at small lags (previous slide)

MFM_ch5_innovations

Diagnostics in Other Markets: a Decision Table

Market	Residual pattern	Possible failure	Next check
BET	$Q(10)$ on $\hat{z}$ = 116.6	autocorrelation in the mean (thin trading, Chapter 2)	AutoRegressive model AR(1) for the mean
Bitcoin	$Q(10)$ on $\hat{z}^2$ = 4.7, joint sign bias 1.3: no rejection	$\hat{\nu} = 3.18 < 4$: $\mathrm{Var}(z_t^2) = \infty$, so the χ^2 calibration is not justified	tests on $\lvert\hat{z}_t\rvert$
EUR/RON	excess kurtosis 3 190	one residual of 154 (May 2025) dominates	inspect the largest residuals; a bounded-score model
Gold	EGARCH: $Q(10)$ on $\hat{z}^2$ = 48.8	clustering left despite a good BIC	choose by diagnostics, not by BIC alone

- Always look at the largest residuals: one observation can dominate every diagnostic

MFM_ch5_innovations

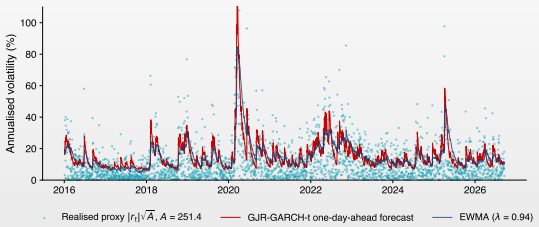
The Proxy Problem and Robust Losses

- **The question**: do the BIC gains of GJR and EGARCH survive out-of-sample?
- Volatility is never observed: we compare forecasts with a noisy **proxy**; here r_t^2, since we use no intraday data (realised variance: Chapter 9)
- $r_t^2 = \sigma_t^2 z_t^2$ is unbiased ($\mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$ for $\mu = 0$) but very noisy
  - Even the perfect forecast $h_t = \sigma_t^2$ explains little of the variation of r_t^2 (a small R^2), because z_t^2 is noisy Andersen & Bollerslev (1998)
  - S&P 500: we ignore the mean, since $\hat{\mu}^2 = 0.0062$ against an average daily variance of 1.47 ($\%^2$), a bias below 1%
- Patton (2011): only some loss functions rank forecasts correctly with a noisy proxy
  - Robust: Mean Squared Error (MSE) $(r_t^2 - h_t)^2$ and **Quasi-Likelihood (QLIKE)** $r_t^2/h_t + \ln h_t$, with h_t the variance forecast for day t
  - Not robust: $(|r_t| - \sqrt{h_t})^2$ and $(\ln r_t^2 - \ln h_t)^2$: their ranking depends on the proxy noise
- QLIKE is (up to constants) minus the Normal log-likelihood; its expected value is lowest at $h_t = \sigma_t^2$; it penalises under-prediction more

Forecast Evaluation: Mincer–Zarnowitz and Diebold–Mariano

- **Mincer–Zarnowitz (MZ) regression** Mincer & Zarnowitz (1969): regress the proxy on the forecast
  - $r_t^2 = a + b\, h_t + u_t$
- Unbiased forecast: $a = 0$, $b = 1$
  - Joint Wald test with Heteroskedasticity and Autocorrelation Consistent (HAC) errors; χ^2_2 under H_0
  - $b < 1$: forecasts too dispersed (too high when high, too low when low)
- **Diebold–Mariano test** Diebold & Mariano (1995): do two forecasts have the same expected loss?
  - Loss differential $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$
  - L: loss function (QLIKE or MSE); h_t^A, h_t^B: forecasts of models A and B; $d_t < 0$: A did better on day t
- Diebold–Mariano (DM) statistic: $DM = \bar{d} / \widehat{\mathrm{SE}}(\bar{d})$
  - $\bar{d}$: mean of d_t over the forecast days; HAC standard error, since d_t is autocorrelated
  - $\approx N(0,1)$ under equal accuracy; $DM < -1.96$: A significantly better; $DM > 1.96$: B significantly better
- The $N(0,1)$ law assumes given forecasts; for estimated, nested models (GARCH-t inside GJR-t) it can fail, so DM t is read descriptively there (Giacomini–White, GW, below)

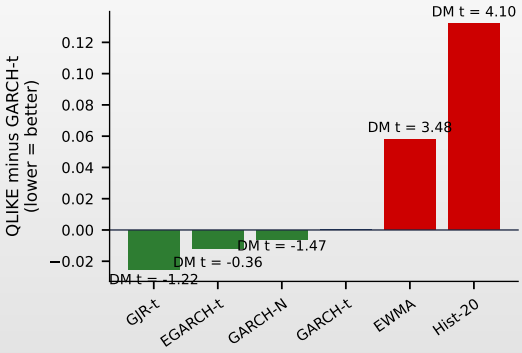
Out-of-Sample Design and Forecast Paths, 2016–2026



- S&P 500: 2 692 one-day forecasts; refit each January on all data since 2000 (expanding window), daily updates with fixed parameters
- Timing: h_t uses data up to the close of day $t-1$ and is compared with r_t^2; changing r_t must leave h_t unchanged
- Models: GARCH-N, GARCH-t, GJR-t, EGARCH-t
- Benchmarks without estimated parameters: EWMA ($\lambda = 0.94$) and Hist-20, the mean of r^2 over the last 20 days

MFM_ch5_forecast_evaluation

S&P 500: Does Anything Beat GARCH(1,1)?



- Mean QLIKE minus that of GARCH-t, with the DM t (HAC, 5 lags); lowest QLIKE: GJR-t (0.732)
- GARCH family: DM t from -1.47 to -0.36, descriptive for these nested pairs
- EWMA and Hist-20 are clearly worse ($t = 3.48$, 4.10): mean reversion and estimated parameters pay off Hansen & Lunde (2005)
- With MSE the ranking changes: EGARCH-t 21.30, GJR-t 21.81, GARCH-N 22.28, GARCH-t 22.76 ($\%^4$)
- MZ: unbiasedness not rejected for GARCH-type forecasts, rejected for Hist-20 (all statistics: Appendix)
- Not rejecting is weak evidence: SE of b is 0.25 for GARCH-t, and a few crisis days drive b (Seminar 5, B5)

MFM_ch5_forecast_evaluation

Comparing Forecasting Methods (1/3): the Limit of DM and the Giacomini–White Test

- **The question**: how do we compare forecasts of nested models with estimated parameters?
- **The limit of the DM test** West (1996), Clark & McCracken (2001): with estimated parameters, nested models and an expanding window, the loss differential degenerates under H_0, and DM is no longer $N(0,1)$
- **The Giacomini–White test** Giacomini & White (2006) tests the forecasting **method**: the model plus a rolling window of $W = 1\,000$ days
- The estimation noise does not vanish, so the test stays valid for nested models
- $GW = n\,\bar{Z}^\top \hat{\Omega}^{-1} \bar{Z} \to \chi^2_2$ under H_0: $\mathbb{E}[d_{t+1} \mid \mathcal{F}_t] = 0$, with $Z_{t+1} = (1, d_t)^\top d_{t+1}$
  - d_{t+1}: loss differential; $\bar{Z}$: mean of Z_{t+1}; $\hat{\Omega} = n^{-1} \sum Z_{t+1} Z_{t+1}^\top$: the uncentred second moment; n: number of forecasts
- Reject at 5% if $GW > 5.99$

Comparing Forecasting Methods (2/3): the Model Confidence Set

- **The question**: which models remain candidates for "the best" when we compare them all?
- With six models, the comparisons with GARCH-t are five tests; the **Model Confidence Set (MCS)** Hansen, Lunde & Nason (2011) compares all 15 pairs
- Multiple tests raise the rate of false rejections
- The MCS drops the worst model until equal accuracy is no longer rejected
- The final set contains the best model with probability $\geq 1 - \alpha$, here $\alpha = 10\%$
- Stationary bootstrap: resamples blocks of random (geometric) length, mean 10 days, so that serial dependence is kept

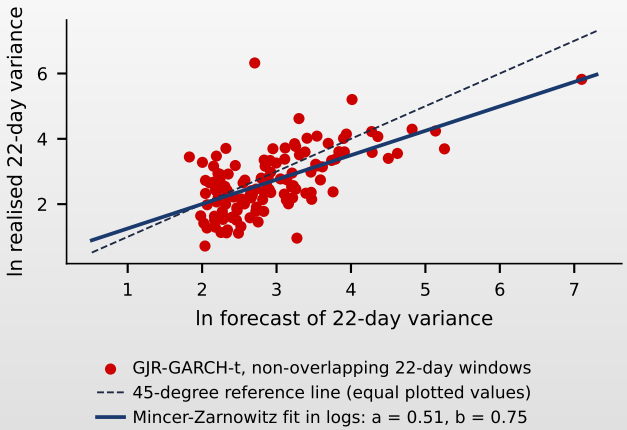
Comparing Forecasting Methods (3/3): Results for the S&P 500

- **Giacomini–White**: rolling window, 2 692 forecasts, each model against GARCH-t
  - Rejects only EWMA and Hist-20
- **Model Confidence Set (MCS) 90%**: expanding window, QLIKE, stationary bootstrap
  - Keeps the four GARCH models, eliminates EWMA and Hist-20
- Same conclusion as the DM test; GW, with a rolling window, has calibrated inference, while the MCS (expanding window) carries the same caveat as DM (Full table and the MCS statistic: Appendix)

Model	GW p	MCS p
GARCH-N	0.054	0.586
GARCH-t	–	0.353
GJR-t	0.288	1.000
EGARCH-t	0.201	0.586
EWMA	0.003	0.008
Hist-20	< 0.001	0.001

MFM_ch5_inference

One-Month Forecasts: Mincer–Zarnowitz in Logs



- GJR-t forecast of the 22-day variance against the realised sum of 22 squared returns, 122 non-overlapping windows; regression $\ln Y = a + b \ln \hat{h} + u$, Y: realised sum, $\hat{h}$: forecast; R^2 in logs 0.38
- Slope in logs $b = 0.75$ (SE 0.07): descriptive only, since $\mathbb{E}[\ln Y \mid \mathcal{F}] \neq \ln \mathbb{E}[Y \mid \mathcal{F}]$; $b < 1$ does not prove over-dispersed forecasts
- In levels b is only 0.27: two COVID-19 windows dominate
- Hence we compare the 22-day forecasts with QLIKE Patton (2011): GARCH-N 4.20, GARCH-t 4.21, GJR-t 4.24, EWMA 4.36 (lower = better)
- At one month, the GJR asymmetry no longer pays off

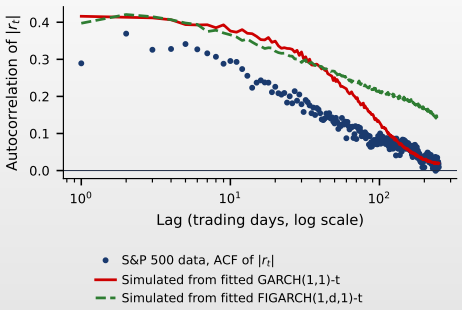
MFM_ch5_forecast_evaluation

GARCH-in-Mean: Is Risk Rewarded?

- **The question**: what does GARCH(1,1) miss, and which extension answers each limit?
- **GARCH-in-Mean (GARCH-M)** Engle, Lilien & Robins (1987): $r_t = c + \kappa\,\sigma_t + \varepsilon_t$ (or $\kappa\,\sigma_t^2$)
  - c: constant; κ: price of risk, the extra expected return per unit of volatility
  - $\kappa > 0$: investors demand a higher expected return when risk is higher (risk-return trade-off)
- S&P 500, GARCH(1,1)-t in mean: $\hat{\kappa} = 0.096$ (t = 3.02) with σ_t; 0.028 (t = 2.35) with σ_t^2
  - Positive and significant in this sample, but the literature is mixed: Glosten, Jagannathan & Runkle (1993) found a negative relation
- A daily mean effect is small and fragile; it is sensitive to the sample and to the asymmetry in variance

MFM_ch5_long_memory

Long Memory: the Shape of the ACF of $|r_t|$



- S&P 500, ACF of $|r_t|$ (log scale): in the data, 0.289 at lag 1, 0.082 at lag 100 and 0.009 at lag 250
- A simulated GARCH(1,1)-t overstates the ACF at every lag: 0.416 at lag 1, 0.130 at lag 100, 0.020 at lag 250
- The problem is the shape: a single $\alpha + \beta$ cannot give both a low level at lag 1 and a slow decay
- **Fractionally Integrated GARCH (FIGARCH)** Baillie, Bollerslev & Mikkelsen (1996): the filter $(1 - L)^d$, with $Lx_t = x_{t-1}$ and $d \in (0, 1)$, gives a hyperbolic decay
- $\hat{d} = 0.60$ (SE 0.06), BIC lower by 34.2
- FIGARCH overstates the memory even more: 0.215 at lag 100, 0.144 at lag 250
- No single-scale model fits all horizons: realised volatility and Heterogeneous AutoRegressive (HAR) models in Chapter 9

MFM_ch5_long_memory

Limits of GARCH and the Extensions They Motivate

Limit	Evidence in this chapter	Extension
Level shifts inflate $\alpha + \beta$ Lamoureux & Lastrapes (1990)	BET-TR (2014–2026) far less persistent than the BET (2000–2026): a hint, not proof, as sample and index differ	GARCH-MIDAS (Mixed Data Sampling), a slow long-run component Engle, Ghysels & Sohn (2013); regime switching Hamilton & Susmel (1994)
Reaction lag of one day	COVID-19: the VIX moved the same day, GARCH the next	Realized GARCH, with a realised-variance measurement equation Hansen, Huang & Shek (2012) (Chapter 9)
One outlier drives the update	EUR/RON residual of 154 (May 2025)	GAS (Generalised Autoregressive Score) models, bounded score Creal, Koopman & Lucas (2013) (case study below)
Shape of the ACF of $\lvert r_t \rvert$	GARCH-t and FIGARCH overstate the ACF (previous slide)	HAR on realised variance (Chapter 9)
One asset at a time	portfolios need covariances	Multivariate GARCH and DCC, Dynamic Conditional Correlation (Chapter 6)
Managed prices	EUR/RON: long calm, then jumps	Model interventions as jumps or regimes

- Each extension is judged like GARCH: out-of-sample, robust loss, Model Confidence Set (model equations: ▸ Appendix)

MFM_ch5_garch_estimation

Two Research Extensions and What They Assume

- **Quasi score-driven volatility**: a replication of a 2023 journal article
  - Builds on: the Student-t density, LR tests and their boundary versions, the score-driven (GAS) update of the extensions section
  - Continued in Seminar 5, C3
- **AI for scientific discovery**: an assistant-driven research loop with checks
  - Builds on: the out-of-sample design, QLIKE, Diebold–Mariano and the Model Confidence Set
  - Its checklist supports the AI audit of Seminar 5, C1 and the team project

Score-Driven Volatility: Should Shocks Be Downweighted?

- **Question** Blasques, Francq & Laurent (2023)
  - Must a score-driven volatility model update σ_t^2 with the score of the same density that describes the returns?
- Score-driven (GAS) update Creal, Koopman & Lucas (2013): tomorrow's parameter moves in the direction that would have made today's observation more likely
  - $f_{t+1} = \omega + \alpha\, S(f_t)\, \partial \log p(y_t \mid f_t)/\partial f_t + \beta f_t$
  - y_t: the return; f_t: the time-varying parameter (here the variance); $p(y_t \mid f_t)$: the density of y_t given f_t
  - $\partial \log p/\partial f_t$: the score; $S(f_t) > 0$: a scaling factor; ω, α, β: as in GARCH
- With a Student-t density, the same parameter ν sets both the tail thickness of z_t and how strongly a large shock is downweighted in the update

QSD: Separating Tail Thickness from Shock Downweighting

- **Quasi Score-Driven (QSD) GARCH(1,1)-t**, eq. (30) of the paper: $y_t = \sqrt{f_t}\,\epsilon_t$, ϵ_t standardised Student-t with $1/\xi$ degrees of freedom (f_t is σ_t^2, ϵ_t is z_t)
  - $f_{t+1} = \omega + \varpi X_t + \alpha \dfrac{1+\zeta}{1-2\zeta+\zeta\epsilon_t^2}\,\epsilon_t^2 f_t + \beta f_t$
  - X_t: an exogenous variable, the squared VIX at a daily horizon; ϖ: its coefficient
  - ξ: the tail thickness of ϵ_t ($\xi = 0$: Normal); ζ: the downweighting of large ϵ_t^2 in the update
  - $\zeta = 0$: GARCH(1,1)-t; $\zeta = \xi$: Beta-t-GARCH (score-driven)
  - Why: with $\zeta = \xi = 1/\nu$, multiplying by ν gives $\dfrac{1+\zeta}{1-2\zeta+\zeta\epsilon_t^2} = \dfrac{\nu+1}{\nu-2+\epsilon_t^2}$, the weight in the Student-t score
- Data: 400 S&P 500 stocks, 3 January 1995 (or later) – 28 February 2019; maximum likelihood with an AR(1) mean
  - LR tests of $\zeta = 0$ and $\xi = \zeta$ against χ_1^2 (full method: Appendix)
  - Probability Integral Transform (PIT) test: $u_t = F(y_t \mid \mathcal{F}_{t-1})$, the fitted distribution function at the observed return, must be i.i.d. uniform on $(0,1)$ if the density is right

QSD: Key Results (Paper, Tables 4 and 6)

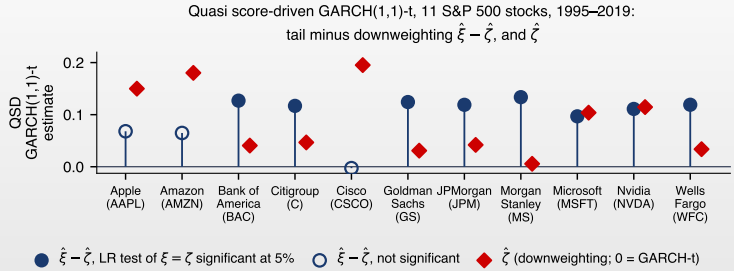
Rejected at 5%, % of the 400 stocks	Student-t (Table 4)	Skewed t (Table 6)
LR, H_0: $\zeta = 0$ (GARCH update)	89.5	98.8
LR, H_0: $\xi = \zeta$ (score-driven; skewed: also $\kappa = \tau$)	53.5	100.0
PIT test: GARCH(1,1)	40.0	33.1
PIT test: Beta-t-GARCH	46.0	72.2
PIT test: QSD GARCH(1,1)	45.7	11.3

- κ: asymmetry of the density; τ: asymmetry of the update; $\kappa = \tau = 0.5$: symmetric (skewed-t version, Table 6)
- Downweighting large shocks matters (Table 4)
  - $\zeta = 0$ is rejected for most stocks (first row)
  - Every $\hat{\zeta}$ is positive

QSD: Reading the Results (Paper, Figs. 2–6)

- **Beta-t-GARCH downweights too much** (Figs. 2–3)
  - Where $\hat{\xi} - \hat{\zeta}$ is significant, $\hat{\xi} > \hat{\zeta}$ for all stocks but one
  - Median $1/\hat{\zeta}$: 9.48 in QSD, against 5.07 in Beta-t-GARCH, where $\zeta = \xi$
  - Tail thickness and shock downweighting are two different parameters
- **Skewed t** (Figs. 5–6)
  - $\hat{\tau}$ far below 0.5 (median 0.333), while $\hat{\kappa}$ stays near 0.5
  - Leverage enters through the update equation, not through the density
- **Calibration** (Table 6)
  - With skewed-t updates, the PIT test rejects the QSD model for only 11.3% of the stocks, against 72.2% for Beta-t-GARCH

Case Study: Replication on 11 Stocks (cf. Paper, Fig. 2)



- Circles: $\hat{\xi} - \hat{\zeta}$ (tails minus downweighting) for each stock; filled if the LR test rejects $\xi = \zeta$ at 5%
- Diamonds: $\hat{\zeta}$, the downweighting of large shocks ($\zeta = 0$: GARCH-t)
- $\hat{\xi} > \hat{\zeta}$ for 10 of 11 stocks; the exception is Cisco ($\hat{\xi} = 0.193$, $\hat{\zeta} = 0.195$)

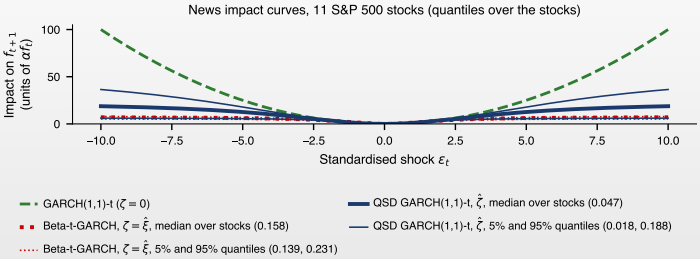
MFM_ch5_quasi_score_driven

QSD: Course Sample and Results

- Course sample: 11 large stocks (the paper: 400), 1995 (or later) – February 2019, 4986–6081 returns each; $X_t = \mathrm{VIX}_t^2/252$
- $\zeta = 0$ rejected for 9 of 11 (paper: 89.5%); $\xi = \zeta$ rejected for 8 of 11 (paper: 53.5%)
- Median $1/\hat{\zeta}$: 21.4 (QSD) against 6.32 (Beta-t-GARCH); paper: 9.48 against 5.07; 11 selected survivors: imprecise and not representative

MFM_ch5_quasi_score_driven

QSD: News Impact Curves (cf. Paper, Fig. 3)



- Updating functions at quantiles of the 11 estimates, at a common αf_t: the contribution of ϵ_t to $f_{t+\mathbf{1}}$, in units of αf_t (GARCH-t adds $\epsilon_t^{\mathbf{2}}\, \alpha f_t$)
- A shock of 5 standard deviations adds 25 (GARCH-t), 12.6 (median QSD), 6.2 (median Beta-t-GARCH); at 10 standard deviations: 100, 18.8, 7.0 (bound $(1+\zeta)/\zeta = 7.3$)
- The QSD downweighting lies between GARCH-t and Beta-t-GARCH

MFM_ch5_quasi_score_driven

Open Questions and the Discovery Loop (1/2)

- Questions
  - do machine-learning or realised-measure forecasts beat a well-specified GARCH for daily equity variance, out of sample?
  - does the answer survive once the number of models compared is accounted for?
- Literature
  - Christensen, Siggaard & Veliyev (2023): machine-learning methods with little tuning are competitive with HAR-type models
  - Hansen, Lunde & Nason (2011): the Model Confidence Set is the right object when many models compete
- Formal claim: $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$, $H_0 \colon \mathbb{E}d_t = 0$
  - Falsified if A leaves the Model Confidence Set
  - Or if the sign of $\bar{d}$ flips across pre-registered subperiods

Open Questions and the Discovery Loop (2/2)

- The loop
  - An Artificial Intelligence (AI) assistant drafts the literature map, competing hypotheses, code, a robustness plan and a referee-style critique
  - The researcher checks every Digital Object Identifier (DOI, a permanent publication identifier), reproduces every number and fixes the plan before the first run
- Checks specific to volatility forecasts
  - Look-ahead: h_t must not move when r_t changes
  - A proxy-robust loss (QLIKE, MSE)
  - Model mining: Model Confidence Set or a Bonferroni correction
  - Scale and convergence of the optimiser
  - DOIs or `arch` function arguments invented by the assistant (reusable prompts and the full checklist: Appendix)

Mini-Case and a Team-Project Seed

- A typical assistant claim: "GJR-t beats GARCH-t because of the leverage effect"
  - d_t = QLIKE(GJR-t) − QLIKE(GARCH-t), 2 692 days, 2016–2026: $\bar{d} = -0.025$, DM $t = -1.22$
- Testable implication of leverage: the gain sits after down days
  - After $r_{t-\mathbf{1}} < 0$ (1 223 days): $\bar{d} = -0.015$, $t = -0.51$; after $r_{t-\mathbf{1}} \geq 0$ (1 469): $\bar{d} = -0.034$, $t = -1.12$
  - Not supported; by period: 2016–2020 $t = -0.02$, 2021–2026 $t = -4.06$
  - Descriptive only (nested models, expanding window); calibrated follow-up: a fixed-window GW test with d_t regressed on $1(r_{t-\mathbf{1}} < 0)$
- **Project seed**: do machine-learning forecasts enter the Model Confidence Set together with GARCH, GJR and EGARCH for the S&P 500 and the BET, 2016–2026?
  - Replicate the QLIKE table first; pre-register models, losses, splits and error rate
  - Log AI use in `AI_USE.md` and every caught error in `AI_ERRORS.md` (full brief: Appendix)

Synthesis: a Decision Workflow

- **1. Detect**
  - $Q(10)$ on r_t^2 and the LM test
  - Clustering appears in all six markets; persistence is close to 1, and for Bitcoin and EUR/RON the fit hits the IGARCH bound
- **2. Fit**
  - GARCH, GJR, EGARCH with Normal, Student-t, skewed-t innovations; returns in %
  - Check convergence and parameters at a bound
- **3. Diagnose**
  - Level, squares and signs of $\hat{z}_t$, and its largest values
  - Naive tests are conservative
- **4. Forecast**
  - The recursion gives multi-step variances
  - The square-root-of-time rule ignores mean reversion
- **5. Compare**
  - Out-of-sample: QLIKE, GW with a rolling window and the Model Confidence Set
  - GARCH beats EWMA and Hist-20; the GARCH family is hard to separate
- **6. Report uncertainty**
  - Robust SE (valid if $\mathbb{E}z_t^4 < \infty$) and boundary-aware tests
  - Half-lives as intervals $[h_{lo}; \infty)$ when $\alpha + \beta \approx 1$

Check Yourself, and Next: Chapter 6

Retrieval questions

- Why does the square-root-of-time rule overstate the 10-day risk after a turbulent day?
- Why is the LR test of Normal against Student-t innovations compared with 2.71 and not 3.84?
- Why can a symmetric interval for the half-life contain negative values?
- Why is a gap between the VIX and GARCH volatility not a measured variance risk premium?

Next chapters

- Chapter 6: multivariate volatility and dynamic correlations (DCC)
- Chapters 7–8: Value at Risk and expected shortfall with GARCH volatility, and their backtesting
- Chapter 9: realised volatility and long memory
- Seminar 5: the variance recursion step by step, GARCH-t and GJR on real data, an AI answer to audit, bootstrap tests and forecast comparisons

Check Yourself: Short Answers

- **Square-root-of-time rule**
  - After a turbulent day $\sigma^2_{t+1} > \sigma^2$, the long-run variance; the daily forecasts decay towards σ^2, so their sum is below $H\sigma^2_{t+1}$
  - Seminar 5, A1
- **2.71 instead of 3.84**
  - Under H_0, $1/\nu = 0$ is on the boundary: in about half of the samples the estimate sits at the bound and $LR = 0$, so the law is $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$, whose 5% critical value is 2.71
- **Negative half-lives**
  - $h = \ln 0.5 / \ln p$ explodes as $p \to 1$; the delta method replaces it by a straight line, so $\hat{h} \pm 1.96\,\widehat{\mathrm{SE}}$ ignores the curvature; map the interval for p instead
  - Seminar 5, A2
- **VIX against GARCH**
  - The right comparison: VIX^2, the 30-day risk-neutral expectation of variance, against a 30-day GARCH variance forecast under the real-world probability; $\hat{\sigma}_t$ is a one-day volatility, so horizon and probability both differ

Data Used in This Chapter

Series	Owner / source	Start	Obs.	Obs. per year	Vol. (% p.a.)	Excess kurt.
S&P 500	S&P Dow Jones Indices	2000	6 714	251.4	19.2	10.7
BET	Bucharest Stock Exchange	2000	6 670	249.8	22.1	11.1
BET-TR	Bucharest Stock Exchange	2014	2 992	249.7	15.4	17.8
Bitcoin (BTC/USD)	crypto exchanges (composite)	2014	4 384	365.3	66.6	12.0
EUR/RON reference rate	BNR	2005	5 352	252.4	4.4	15.1
Gold (XAU/USD)	spot market	2000	6 922	259.2	17.6	7.7

- Equity indices and gold: weekdays only
- Vol.: sample volatility annualised with each series' observations per year

MFM_ch5_arch_effects

Back

One-Sided Score Tests and the Davies Problem

- The alternative is one-sided ($\alpha_i \geq 0$): the two-sided χ^2 test wastes power
  - Lee & King (1993): one-sided score test built on the sum of the score components, $N(0,1)$ under H_0, rejects in the upper tail only
- Against GARCH(1,1) the null $\alpha = 0$ leaves β unidentified (Davies problem)
  - The ARCH(q) LM statistic is also the score test against GARCH(p,q): the same statistic serves both alternatives (Lee & King (1993))

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Robert Engle and the 2003 Nobel Prize



Robert F. Engle (2022)

Photo: Jon Demske (2022); CC BY-SA 4.0; Wikimedia Commons



Clive W.J. Granger (2008)

Photo: Olaf Storbeck (2008); CC BY-SA 2.0; Wikimedia Commons

- Nobel Prize 2003: Engle "for methods of analyzing economic time series with time-varying volatility (ARCH)"
  - Shared with Clive Granger (cointegration)
- Nobel lecture: Engle (2004), on risk and volatility in financial practice
- Tim Bollerslev, Engle's doctoral student, generalised ARCH to GARCH in 1986

Back

Kurtosis: Rounding and a Long Simulated Path

- S&P 500, GARCH-N: the moment condition is 0.990435, so $K = 12.1$ has the denominator $1 - 0.990435$
  - Rounding to $\hat{\alpha} = 0.121$, $\hat{\beta} = 0.860$ gives 0.9916 and $K = 13.5$: report such functions from unrounded estimates
- Simulated GARCH(1,1) of the simulation slide, one million days
  - Excess kurtosis 2.47 (theory 3.06); lag-1 autocorrelation of ε_t^2: 0.26
  - The sample kurtosis converges slowly: its precision depends on the eighth moment, infinite for these parameters

Back

QMLE: Variants of the Theorem

- Estimated mean: the score gains the term $\varepsilon_t \, (\partial \mu_t / \partial \theta) / \sigma_t^2$
- The covariance is no longer $(\kappa_z - 1) J^{-1}$: the mean block also depends on the skewness of z_t
- IGARCH ($\alpha + \beta = 1$): strict stationarity still holds, and the QMLE remains consistent and asymptotically Normal Lumsdaine (1996)
- Explosive case, $\mathbb{E} \ln(\alpha z_t^2 + \beta) > 0$: Jensen & Rahbek (2004) estimate (α, β) with ω fixed, since ω cannot be estimated consistently

Back

The Li–Mak Correction for Squared-Residual Autocorrelations

- Squared-residual autocorrelations $\hat{r}_k$: $\sqrt{n}\,\hat{r} \to N(0, C)$, $C = I - H' J^{-1} H/(\kappa_z - 1)$ Li & Mak (1994)
  - $\hat{r} = (\hat{r}_1, \ldots, \hat{r}_m)'$; I: identity matrix; C: the corrected covariance, smaller than I on the diagonal; g_t, J: as in the QMLE theorem
  - $H_k = \mathbb{E}[g_t(z_{t-k}^2 - 1)]$ (k-th column of H): how the fitted variance absorbs lag-k dependence; the ARMA rule χ^2_{m-p-q} does not apply
- S&P 500, Gaussian QMLE of GARCH(1,1): $C_{1,1} = 0.78$, $C_{5,5} = 0.93$, $C_{10,10} = 0.97$
  - The correction is largest at small lags, where $\hat{\alpha}$ has already fitted the dependence
  - The corrected statistic $n\hat{r}' C^{-1} \hat{r}$ is χ^2_{10} under H_0

MFM_ch5_inference

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S&P 500 Forecast Evaluation, All Statistics

Model	QLIKE	MSE	DM t vs GARCH-t	MZ b	MZ R^2	Wald p
GARCH-N	0.751	22.3	−1.47	0.97	0.250	1.00
GARCH-t	0.757	22.8	–	0.88	0.238	0.80
GJR-t	0.732	21.8	−1.22	0.84	0.276	0.68
EGARCH-t	0.745	21.3	−0.36	1.24	0.293	0.71
EWMA	0.815	24.7	3.48	0.89	0.172	0.69
Hist-20	0.889	26.1	4.10	0.68	0.152	0.03

- Mean losses on 2 692 days; DM on QLIKE, HAC with 5 lags; negative DM t: the model beats GARCH-t
- MZ: $r_t^2 = a + b\,h_t$; Wald p: test of $a = 0$, $b = 1$

MFM_ch5_forecast_evaluation

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Giacomini–White and the MCS: Full Table and Settings

Model	QLIKE (W)	t uncond.	GW	p	MCS p
GARCH-N	0.751	1.87	5.85	0.054	0.586
GARCH-t	0.743	–	–	–	0.353
GJR-t	0.738	−0.18	2.49	0.288	1.000
EGARCH-t	0.746	0.07	3.21	0.201	0.586
EWMA	0.815	3.20	11.40	0.003	0.008
Hist-20	0.889	3.87	16.41	< 0.001	0.001

- GW: the statistic of the main text, χ^2_2 under H_0 (5.99 at 5%); t uncond.: the unconditional test (instrument 1 only), a Diebold–Mariano t on the rolling-window losses
- Rolling window $W = 1\,000$ days, re-estimated each January, 2 692 forecasts; tests against GARCH-t: unconditional t uses HAC with 5 lags; GW uses the uncentred second moment, without HAC
- MCS on the expanding-window forecasts: QLIKE, statistic T_R (the largest pairwise $|t|$ of loss differences), stationary bootstrap (mean block 10, 10 000 draws)

MFM_ch5_inference

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Score-Driven, Realized and Component Models

- **Score-driven (GAS)** Creal, Koopman & Lucas (2013): $\sigma^2_{t+1} = \omega + \alpha\, s_t + \beta\, \sigma^2_t$, s_t the scaled score of the Student-t density with respect to σ^2_t
  - The score is bounded in z_t: an outlier such as the EUR/RON residual of 154 moves the variance far less than in GARCH
- **Realized GARCH** Hansen, Huang & Shek (2012): adds a measurement equation linking σ^2_t to a realised variance
  - Uses intraday information and reacts faster; the bridge to Chapter 9
- **GARCH-MIDAS** Engle, Ghysels & Sohn (2013): $\sigma^2_t = \tau_t\, g_t$, a slow long-run component τ_t driven by macro data (mixed frequencies)
  - g_t: a unit-mean GARCH(1,1) component for the short-run moves
  - A moving long-run level absorbs the level shifts that push $\hat{\alpha} + \hat{\beta}$ towards 1

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Case Study, Full Method (1/2)

- **Why it matters**: GARCH is driven by ε_t^2 whatever the density, so an isolated extreme shock can produce an excessive volatility response lasting days or weeks (Section 1 of the paper)
  - With a left-skewed density, a pure score-driven model raises volatility more after positive returns: the opposite of the leverage effect (Example 2)
  - QSD models, eq. (3): $f_{t+1} = \omega + \alpha\, \psi(y_t, X_t, f_t, \theta) + \beta f_t$, ψ the scaled score of an updating function ρ that need not equal $\log p$; theory: consistency, asymptotic normality, LR and LM tests with χ^2 limits (Section 3)
- **Estimation**: $\zeta = 0$ is kept inside the parameter space by eqs. (31)–(32): $\zeta \in (-1, 1/2)$ and $\Psi(x) = x\,(1 - e^{-cx})/(1 + e^{-cx})$ applied to the weight, $c = 1000$
  - PIT test Diebold, Gunther & Tay (1998) with the Pierce (1982) correction for parameters estimated by maximum likelihood: reject if $|\sum_t \hat{u}_t - T/2| / \sqrt{T(1/12 - B\,\mathcal{I}(\theta)^{-1}B^\top)} > 1.96$
  - $B = \lim_{T\to\infty} T^{-1}\mathbb{E}[\partial \sum_t u_t / \partial\theta^\top]$, $\mathcal{I}$: the Fisher information per observation, T: the sample length; it detects a shifted PIT mean, not every departure from uniformity

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Case Study, Full Method (2/2)

- **Data**: stocks with fewer than 4000 observations or with convergence problems are dropped: 400 stocks (Section 5)
  - Skewed variant, eq. (47): a skewed Student-t with asymmetry κ for ϵ_t and separate (ζ, τ) in the update; $\tau < 0.5$ gives a leverage effect whatever κ (Section 5.2)

Case Study, an Example and the Limits Stated by the Authors

- Centene, 4330 days: $\hat{\xi} \approx 0.26$ (about 4 degrees of freedom) in all three models, but $\hat{\zeta} = 0.048$ (SE 0.027); LR 8.81 ($p = 0.003$) for $\xi = \zeta$ and 8.42 ($p = 0.004$) for $\zeta = 0$ (Table 5); after a crash day in March–April 2008, GARCH-t volatility stays well above Beta-t-GARCH and QSD lies between (Fig. 4)
- **Limits** stated by the authors
  - Beating GARCH-t and Beta-t-GARCH does not make the Student-t QSD model adequate: the three models fail the PIT test for 40.0–46.0% of the stocks (Table 4)
  - In the unmodified recursion (30), $\zeta < 0$ or $\zeta > 1/2$ does not define a proper volatility model, and $\zeta = 1/2$ with $\varpi = 0$ is degenerate (Section 4.1)
  - The $N(T/2, T/12)$ law of $\sum_t u_t$ holds only for known parameters; with estimated ones the PIT test needs the correction above (Section 5.1)
- **Open extensions** (Seminar 5, C3)
  - Does the bounded QSD update beat GJR-GARCH-t in QLIKE out of sample, especially after March 2020?
  - Does it tame the EUR/RON residual of 154?

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The Discovery Loop: Reusable Prompts (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature**: *"List out-of-sample comparisons of GARCH, HAR and machine-learning variance forecasts since 2015, with DOI, asset, loss function, test and horizon."*
  - Resolve every DOI on Crossref; check that the loss is robust to a noisy proxy Patton (2011)
- **Hypotheses**: *"For the claim that asymmetry improves one-day forecasts, list competing mechanisms (leverage, volatility feedback, a few crisis days) and what each predicts for d_t given the sign of r_{t-1} and by subperiod."*
- **Data and code**: *"Write Python with `arch` that re-estimates GARCH-t and GJR-t each January on an expanding window from the course daily price files and stores one-day variance forecasts; no parameter may use data after the forecast origin."*
  - Replicate first: QLIKE 0.757 (GARCH-t) and 0.732 (GJR-t), DM $t = -1.22$

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The Discovery Loop: Reusable Prompts (2/2)

- **Robustness**: *"Write a pre-registered plan: splits by the sign of r_{t-1} and by subperiod, MSE besides QLIKE, the 22-day horizon, the HAC lag, and a Bonferroni or Model Confidence Set correction."*
  - A random-split placebo is calibrated only if it reassigns blocks of days, preserving volatility clustering
- **Critique**: *"Act as a referee: could the proxy r_t^2, one crisis year or the HAC lag create my result? Give the test that would expose each."*
- **Report**: all splits and losses tried, adjusted p-values, code, seeds and package versions; a specification chosen after seeing results is exploratory and reported as such
- **Influential periods**: a crisis year can dominate $\bar{d}$; report subperiods and state in advance whether any year may be excluded

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The Team-Project Seed in Full

- Realised-variance (HAR) branch: the course data set of 5-minute realised variances holds 40 large American stocks, January 2015 – January 2026; there are no intraday data for the S&P 500 index or the BET, so this branch is run stock by stock
- Replicate first the QLIKE table and the mini-case splits
- Deliverables
  - A one-page pre-registration: models, losses, splits, error rate and what would falsify the claim
  - Code on the course market data, the Model Confidence Set at 10%, a referee-style self-critique; `AI_USE.md` and `AI_ERRORS.md`
- Code of the mini-case: Quantlets/Ch_05/ai_discovery_case.py

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Further Resources

- Books
  - Franke, Härdle & Hafner (2019), chapters on AutoRegressive Integrated Moving Average (ARIMA) and GARCH models for financial time series
  - Francq & Zakoïan (2019): the mathematical theory of GARCH models
- Quantinar courses
  - Statistics of Financial Markets
  - Applied Time Series Analysis with Python
- Code
  - All charts and tables: Quantlets/Ch_05; Python package `arch` by Kevin Sheppard (documentation)
- Related work: entropy as a tail-risk signal beyond variance
  - Pele, Lazar & Dufour (2017): entropy of intraday EUR/JPY returns, Value at Risk and expected shortfall
  - Pele & Mazurencu-Marinescu-Pele (2019): intraday Bitcoin entropy and daily Value at Risk
  - Pele, Lazar & Mazurencu-Marinescu-Pele (2019): tail entropy and expected shortfall

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Acronyms used in this material (1/2)

- **ACF**: AutoCorrelation Function
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **BIC**: Bayesian Information Criterion
- **BTC**: Bitcoin
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **COVID-19**: COronaVIrus Disease 2019
- **DCC**: Dynamic Conditional Correlation
- **DOI**: Digital Object Identifier
- **DM**: Diebold–Mariano (test)
- **EGARCH**: Exponential GARCH
- **EUR**: euro, international currency code
- **EWMA**: Exponentially Weighted Moving Average
- **FIGARCH**: Fractionally Integrated GARCH
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-M**: GARCH-in-Mean
- **GARCH-MIDAS**: GARCH with Mixed Data Sampling (long-run component)

Acronyms used in this material (2/2)

- **GARCH-N**: GARCH with Normal innovations
- **GAS**: Generalised Autoregressive Score (model)
- **GED**: Generalised Error Distribution
- **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **GW**: Giacomini–White (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **IGARCH**: Integrated GARCH
- **LM**: Lagrange Multiplier
- **LR**: Likelihood Ratio (test)
- **MCS**: Model Confidence Set
- **MLE**: Maximum Likelihood Estimation
- **MSE**: Mean Squared Error
- **MZ**: Mincer–Zarnowitz (regression)
- **PIT**: Probability Integral Transform
- **QLIKE**: Quasi-LIKElihood (loss function)
- **QMLE**: Quasi-Maximum Likelihood Estimator
- **QQ**: Quantile–Quantile (plot)
- **QSD**: Quasi Score-Driven (model)
- **RON**: Romanian leu, international currency code
- **SE**: Standard Error
- **USD**: US dollar, international currency code
- **VIX**: Chicago Board Options Exchange Volatility Index
- **XAU**: gold, international currency code (one troy ounce)

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